PLATFORM / ARCHITECTURE / PROVENANCE

COMPANY HUB

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PROPOSED RESEARCH / NOT YET VERIFIED

THE ENGINEERING QUESTION

Can a provenance model that separates verbatim record from derived interpretation keep company knowledge traceable and trustworthy?

PROBLEM DEFINITION AND RESEARCH CONTRIBUTION

A proposed internal intelligence system for a small UK firm that links contracts, meetings, emails and documents so they trace to one another and surface in briefings.

- Use synthetic contracts, meeting notes, emails and documents only. No real company data is needed.
- Role based access tiers decide what each person can see in a briefing and in the underlying record.
- AI assisted and never AI executed. People approve links, merges and any statement leaving the firm.

ARCHITECTURE AND IMPLEMENTATION SPECIFICATION

1 STAGE **Source connectors** → 2 STAGE **Provenance store** → 3 STAGE **Entity resolution** → 4 STAGE **Briefing engine**

- Python backend with an append only record store and a hash chained audit log for tamper evidence.
- Entity resolution across people, organisations and contracts with confidence scores and a human merge queue.
- React frontend with Anthropic API reasoning and a Google Workspace connector on synthetic accounts.

ENGINEERING DECISIONS TO MAKE EXPLICIT

The design review must record data contracts, state transitions, failure handling and why the selected baseline is defensible. Every subsystem should have an interface specification and a reproducible test route.

DESIGN BOUNDARIES AND ASSUMPTIONS

⚠ Tamper evidence shows that a change happened but does not prove the original record was true.

⚠ A real deployment still needs legal review and a data protection impact assessment.

ⓘ Scope is proposed unless explicitly identified as documented existing work. All measured claims require reproducible evidence.

COMPANY HUB / PROTOCOL / ACCEPTANCE / REVIEW

EXPERIMENTS, VERIFICATION & PUBLICATION

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EXPERIMENTAL METHODOLOGY

01 Generate a synthetic firm with linked contracts, meetings, emails and documents plus known planted conflicts and duplicates.

02 Compare plain keyword retrieval with provenance linked briefings on claim traceability and missed obligations.

03 Test access tiers with adversarial queries and measure whether any restricted passage leaks into a briefing.

MEASUREMENT AND ACCEPTANCE EVIDENCE

- Share of briefing claims that resolve to a source passage
- Entity resolution precision and recall on synthetic data
- Tamper detection rate on altered records

REQUIRED REPOSITORY ARTEFACTS

- Provenance data model and access policy
- Synthetic firm dataset with ground truth links
- Briefing traceability and leakage test report

TECHNICAL REVIEW QUESTIONS

Which assumptions could invalidate the result? How do the baselines behave under the same conditions? What is the strongest negative case? Which variables, seeds, workloads, software versions and machine properties must be retained for independent reproduction?

VERIFICATION MAPPING

For each acceptance claim, record the test identifier, input configuration, observed result, pass/fail definition and an independent rerun procedure. Failed or inconclusive cases remain in the report rather than being silently excluded.

PUBLICATION DISCIPLINE

Release original source, configuration files, test harnesses, raw measurements and scripts where rights permit. Distinguish simulation outcomes from physical validation and completed achievements from planned evaluation. Results must not be inferred from targets.